

Healthy Skepticism · Session 2 take-home summary

Fault 2: probability reversal

Whenever something is measured, two probabilities are in play. They sound the same and are not.

What can be measured	What you want to know
$P(\text{observation} \mid \text{condition})$	$P(\text{condition} \mid \text{observation})$
When the condition is there, how often do we see this?	Having seen it, how likely is the condition?

The fault is reading one as the other. The gap between them depends on something the observation never tells you: how common the condition is in the first place.

The lawn: same mornings, two answers

One hundred mornings are recorded. On 20 of them it rained. On 18 of those 20 the lawn was wet. But the lawn was also wet on 12 mornings when it had not rained, from sprinklers and dew, so the lawn was wet on 30 mornings in total.

- **If it rained, how often was the lawn wet?** 18 out of 20, which is 90%.
- **The lawn is wet. How often had it rained?** 18 out of 30, which is 60%.

Same hundred mornings, same 18. Nothing changed except which number went underneath. A probability is one count divided by another, and the reversal changes the denominator.

The 100 mornings are illustrative, constructed for teaching rather than measured.

From the lawn to the clinic

A diagnostic test comes with two advertised numbers, and both are forward conditionals of exactly the kind above.

Advertised	What it actually says	What you wanted
Sensitivity	$P(\text{test positive} \mid \text{have the cause})$	$P(\text{have the cause} \mid \text{test positive})$
Specificity	$P(\text{test negative} \mid \text{no cause})$	$P(\text{no cause} \mid \text{test negative})$

Each reversal is undone by the other number, together with how common the cause is. The two mnemonics in circulation, SnNout and SpPin, are shorthand for this: a very sensitive test that is negative rules out, a very specific test that is positive rules in. Neither mnemonic tells you what an ordinary result means.

Every test has both reversals. We take each one where it does more damage: a screening mammogram is good at finding the cancers that are there, so its positive misleads; a COVID rapid test is good at clearing people who are well, so its negative misleads.

A callback is not a diagnosis

Screening mammography is measured at 86.9% sensitivity and 88.9% specificity. About six women in a thousand have cancer, averaged across ages 40 to 89; it climbs with age, nearer 4 per 1,000 in the fifties and 6 or 7 in the sixties. Follow 10,000 women through one screening round at the observed rates.

Out of 10,000 women screened	Count
Have cancer and are called back	51
Have cancer and are missed	8
Do not have cancer and are called back	1,103
Do not have cancer, not called back	8,838
Total called back	1,154

Of the 59 women with cancer, 51 are caught: that is the 87% sensitivity, and it is the number the test was measured on. But of the 1,154 women called back, 51 have cancer. That is 4.4%.

$P(\text{called back} \mid \text{cancer}) = 87\%$

reversed becomes

$P(\text{cancer} \mid \text{called back}) = 4\%$

Same test, same day, same woman. Unlikely is why she should not panic. Not ignorable is why the biopsy is warranted.

A biopsy recommendation is a different and much stronger signal: its positive predictive value is 25.6%. The callback is a request for more information, not a finding.

Lehman CD et al., Breast Cancer Surveillance Consortium, Radiology 2017;283(1):49-58 (1,682,504 screens, 2007-2013).

Why reasonable people disagree about screening

Of US women aged 50 screened every year for ten years, the estimated outcomes span these ranges.

- 0.03% to 0.32% avoid a breast cancer death.
- 49% to 67% have at least one false alarm.
- 0.3% to 1.4% are overdiagnosed and treated for a cancer that would not have harmed them.

Nobody in this argument disputes the sensitivity, the specificity, or the 4.4%. The disagreement is over how to weigh a false alarm for half of them against a death averted for a fraction of one percent. So the guidance splits: the American College of Radiology recommends annual screening from 40, the USPSTF every two years from 40, and the Nordic Cochrane group and the Swiss Medical Board have argued against running screening programmes at all.

Overdiagnosis evidence is contested: Miller AB et al., Canadian NBSS 25-year follow-up (106 of 484 screen-detected cancers, 21.9%); Gøtzsche PC and Jørgensen KJ, Cochrane 2013;CD001877. An 11.6% callback rate compounded over ten independent screens gives about 71%.

A negative narrows the odds, it does not clear you

COVID rapid antigen tests in symptomatic people are measured at 99.1% specificity and 73.0% sensitivity. Take 10,000 symptomatic people at a pre-test probability of 4 in 10.

Out of 10,000 symptomatic people	Count
Have COVID, test positive	2,920
Have COVID, test negative	1,080
No COVID, test positive	54
No COVID, test negative	5,946
Total negatives	7,026

The specificity is superb: almost nobody healthy is wrongly flagged. But of the 7,026 people told they were negative, 1,080 had COVID. So $P(\text{no COVID} \mid \text{negative})$ is 5,946 out of 7,026, which is 85%.

$P(\text{negative} \mid \text{no COVID}) = 99\%$

reversed becomes

$P(\text{no COVID} \mid \text{negative}) = 85\%$

More than one person in seven who was told they were negative had COVID and went out anyway.

Dinnes J et al., Cochrane 2022;7:CD013705. The 4-in-10 pre-test probability is illustrative, chosen for a symptomatic person during a wave, and is not a measured figure.

Courtrooms and classrooms

Three cases where nobody caught the flip, and what it cost.

R v Deen

A forensic expert told the court that the likelihood of the sample coming from any man other than the accused was one in three million. The judge repeated it. That is a statement about how often the evidence appears among innocent men, and it was heard as the chance that the accused was innocent. The Court of Appeal quashed the conviction. The name for this reading is the prosecutor's fallacy.

The Times, 10 January 1994; the rules for presenting such evidence were later set in Doheny and Adams [1997] 1 Cr App R 369.

Roadside drug field tests

A two-dollar roadside kit turns blue for cocaine. It also turns blue for dozens of other things, which has been known since the 1970s. In Houston and Harris County between 2004 and 2015, blue was taken to mean cocaine. The laboratory checked only afterwards, and found that more than 250 people had already pleaded guilty to possessing nothing.

Houston Forensic Science Center and the Harris County conviction integrity unit, reported by ProPublica and The New York Times Magazine, 2016.

AI writing detectors

The vendors advertise accuracy figures: Turnitin has claimed 98%, Copyleaks 99.1%, GPTZero 99%, Winston AI 99.98%. Every one of those is a statement about machines. It says: if this text came from an AI, we will flag it.

A flagged student needs the other number: of the essays you flag, how many were actually written by an AI? Vanderbilt ran that number. Even at a 1% false positive rate, their 75,000 annual submissions would falsely accuse 750 students. They disabled Turnitin's AI detection in August 2023.

Liang W et al., Patterns 2023: seven detectors applied to 91 human-written TOEFL essays flagged 61.3% as machine-written, and about one in five were flagged by all seven, against near-zero error on essays by native English speakers.

The same flip, with a transcript at the end of it. A number about how often the tool catches machines was read as the chance that a student cheated.

What to ask, and what comes next

- Which way round is this number? Is it about the test, or about me?
- How common is this condition among people like me? Without that, no result can be interpreted.
- If the result is positive, how many people with this result actually have the condition?
- If it is negative, how many people with this result have it anyway?

Next session: arbitrary categories and the Bernoulli fallacy. Why one point of blood pressure can change your label without changing your risk, and why published medical advice keeps reversing itself.

Course materials and contact

Slides, videos and the whitepaper library are at healthy-skepticism.com. Decks live in the student area, behind the password given out in class.

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